Structured questions

|!|BCELB0310000001|!| ★ **Cross topic** **STSE**

The photograph below shows a maize cob. The cob consists of many individual seeds called kernels. Each seed results from a separate fertilization of a male and a female gamete. Some seeds are yellow and some are purple.



(a) Suggest the approximate ratio of purple seeds to yellow seeds on the cob. (1 mark)

_____

(b) (i) State the type of variation shown by the cob. (1 mark)

_____

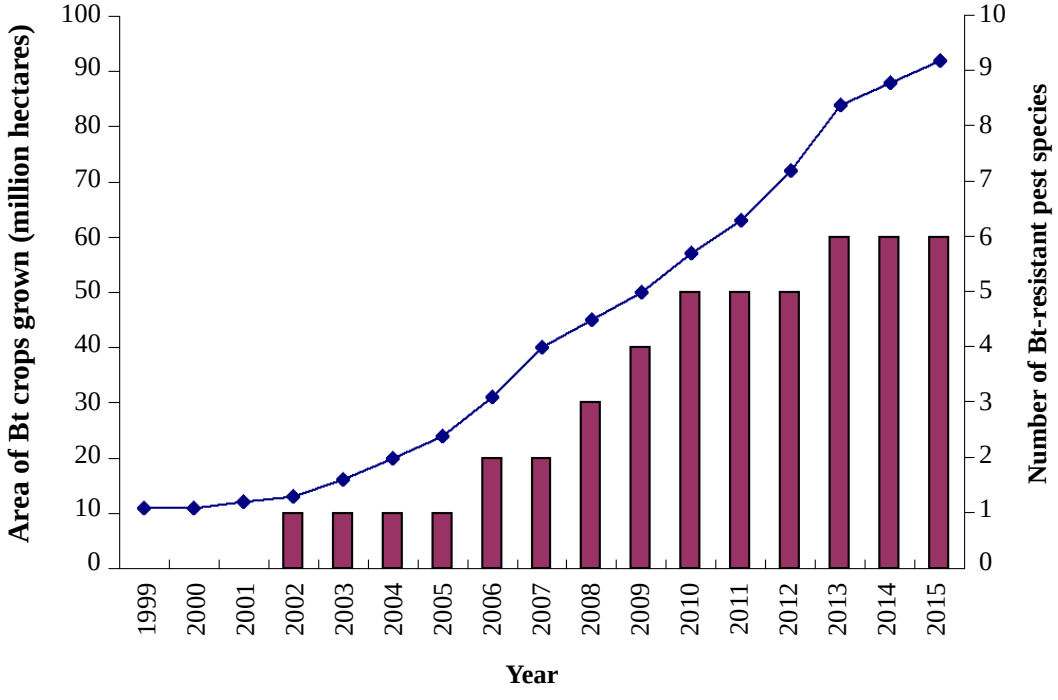
(ii) Using your knowledge about genes and alleles, suggest an explanation for this pattern of variation in seed colour. (2 marks)

_____

_____

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(c) Maize plants have been genetically modified since the 1990s to produce the Bt toxin to kill insect pests. The graph below shows the area of Bt crops grown and the number of Bt-resistant insect pest species.



(i) Describe the relationship between the area of Bt crops grown and the number of resistant pest species. (1 mark)

(ii) Based on the theory of natural selection, suggest an explanation for the relationship in (c)(i). (4 marks)

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##

(a) 1:1 (1)

(b) (i) Discontinuous variation (1)

(ii) Seed colour in maize is controlled by a pair of alleles. (1)

One allele is dominant over the other. (1)

(c) (i) As the areas of Bt crops grown increases, the number of resistant pests species increases. (1)

(ii) Variations in the resistance against Bt toxin exist within populations of insects. (1)

The increased growth of Bt maize favours the survival of resistant insects. (1)

Resistant insects have a greater chance to survive and reproduce. (1)

As a result, the proportion of resistant insects increases within the population. (1)

(9 marks)

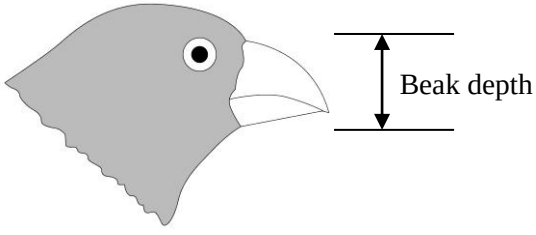
##

	Skills assessed	Marks
(a)	Analysing data	1
(b)(i)	Applying concepts	1
(b)(ii)	Applying concepts	2
(c)(i)	Analysing data	1
(c)(ii)	Applying concepts	4

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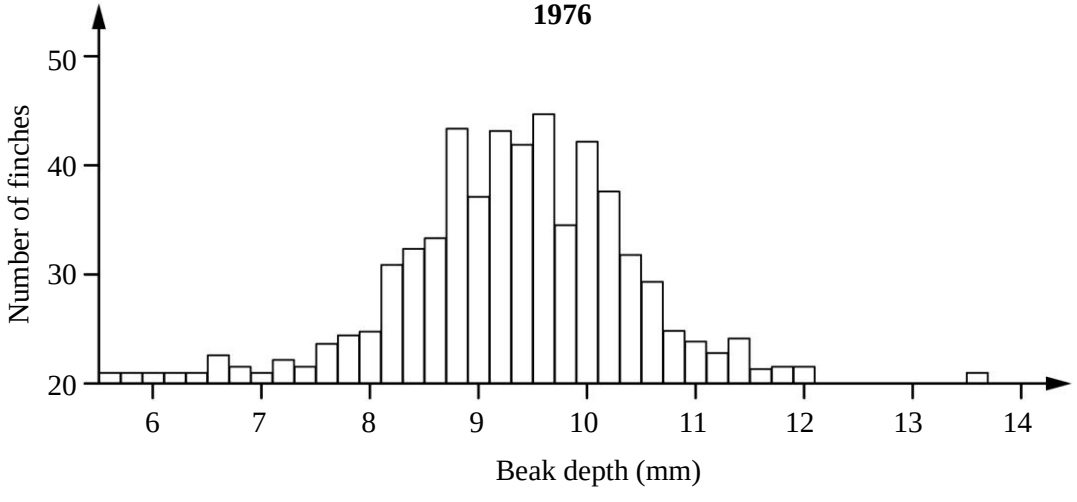
|!|BCELA0310000002|!| ★★

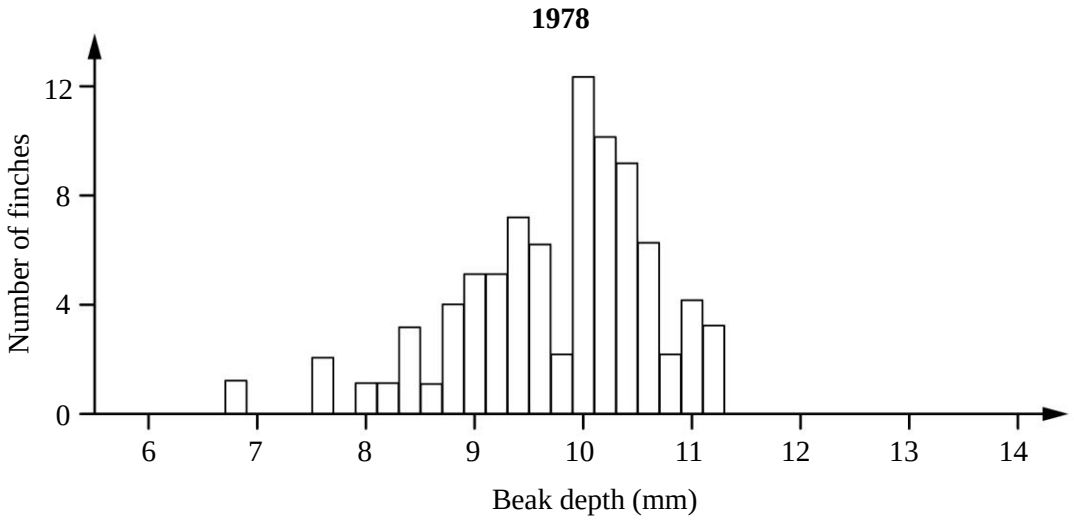
The Galapagos Islands are located about 600 miles from the coast of South America. There are many species of finches living on the islands.
On one of these islands, a population of the seed-eating finch, *Geospiza fortis*, can be found. A group of scientists studied the beak depth of this population of finches. Beak depth is measured as follows.



Finches having large beak depths can eat a wide-variety of seeds including those which are large and hard. When food is abundant, most of the finches choose to eat the smaller, softer seeds. In 1977, there was a severe drought on the island, leading to a decrease in the availability of seeds. The smaller, softer seeds were quickly consumed, leaving only the larger, harder ones.

The following graphs show the beak depth data for the 1976 and 1978 populations of *Geospiza fortis* on the island.





(a) What is the type of variation shown by the beak depth of finches? (1 mark)

(b) (i) According to the graphs above, describe the changes in the finches between 1976 and 1978. (3 marks)

(ii) Explain the changes you have given in (i). (5 marks)

##

(a) continuous variation (1)

(b) (i) The population of the finches decreased in 1978. (1)
The average beak depth of the finches increased in 1978. (1)
The range of beak depth is less diversified in 1978. (1)

(ii) The decrease in food amount could not support the large population of finches, and hence intraspecific competition for food increased. (1)
Those finches that had smaller beaks died out because they could not eat the large, hard seeds. (1)

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Those finches that had larger beaks could eat the large, hard seeds and could survive and reproduce. (1)

The character of having larger beaks could be passed on to offspring. (1)

After many generations, the proportion of finches having larger beaks increased in the population. (1)

(9 marks)

##

	Skills assessed	Marks
(a)	Analysing data	1
(b)(i)	Analysing data	3
(b)(ii)	Applying concepts	5

|!|BCELB0310000003|!| ★ **Cross topic**

There are different varieties of potatoes which are derived from a single species. Some of the varieties are resistant to bacteria. Some are resistant to certain viruses. Some can withstand frost.

(a) Suggest how potato-breeders can clone a particular potato plant with favourable characteristics. (1 mark)

(b) Explain a disadvantage of mass cloning of a particular potato plant. (3 marks)

(c) Suggest how a potato plant which can withstand frost may have evolved in nature. (4 marks)

##

(a) By vegetative propagation (1)

(b) The cloned potatoes are all genetically identical. (1)
This decreases the ability of the clones to adapt to environmental changes / The population of clones lacks genetic variations to adapt to environmental changes. (1)
The clones become more easily attacked by bacteria or viruses. (1)

(c) There is variation among potato plants in their ability to withstand frost. (1)
In a cold environment, potato plants which can withstand frost have selective advantage. (1)
The potato plants which can withstand frost have a greater chance to survive and reproduce than the plants which cannot withstand frost. (1)
The ability to withstand frost can be passed on to offspring. Through many generations, the potato plants which can withstand frost become more frequent in the population. (1)

(8 marks)

##

	Skills assessed	Marks
(a)	Applying concepts	1
(b)	Applying concepts	3
(c)	Applying concepts	4

|!|BCELC0310000004|!| **Cross topic**

The occurrence of mutations is random and mutations can be beneficial, harmful or neutral to organisms. Under the action of natural selection, characteristics favourable for survival in the environment are selected.

(a) Explain what is meant by a mutation. (1 mark)

(b) Mutations can be classified as spontaneous and induced mutations. State ***one*** cause for each type of mutations. (2 marks)

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(c) Explain how mutation could be beneficial to organisms. (3 marks)

(d) Explain how harmful mutations may weaken or kill an organism by affecting its metabolic activities. (3 marks)

##

(a) A mutation is a sudden and permanent change in the DNA of an organism. (1)

(b) Spontaneous mutation arises from errors in DNA replication. (1)
Induced mutation is caused by exposure to mutagens (e.g. X-rays / gamma rays / ultra-violet radiation, etc.). (1)

(c) Mutations can give genetic variations in organisms. (1)
Some genetic variations are favourable for the survival of organisms in their environment. (1)
The individuals with the mutant genes for favourable characters have a greater chance to survive and reproduce. (1)

(d) Mutations may result in the production of abnormal proteins which are not functional. (1)
Hence, mutations may affect enzymes or other proteins that are essential for metabolism. (1)
Metabolic activities are controlled by enzymes or other essential proteins. The lack of these proteins would inhibit metabolic activities. (1)

(9 marks)

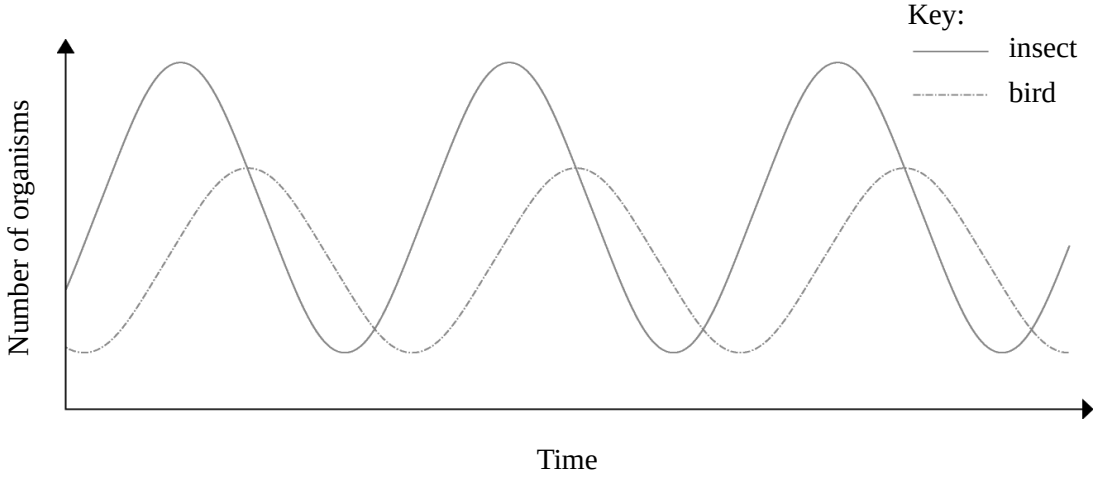
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##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	2
(c)	Applying concepts	3
(d)	Applying concepts	3

|!|BCELB0310000005|!| ★ **Cross topic**

The following graph shows the changes in the numbers of a species of insect and a species of bird in a garden of evergreen trees.



(a) Based on the graph, deduce the ecological relationship between the two species. (3 marks)

(b) With reference to the graph, describe and explain the change in the number of birds when the number of insects decreases. (2 marks)

(c) Some of the insects are brown while some are green colour.

(i) State and explain which body colour of the insects is favourable to survival in this garden. (3 marks)

(ii) Using the concept of natural selection, explain how insects that are green in colour may have become the dominant variety in this garden. (4 marks)

##

(a) Predation (1)
Populations of predators and prey in a community are closely related and fluctuate in a cyclic pattern. (1)
The changes in the predator population always lag behind the changes in the prey population, and the prey population is larger than the predator population. (1)

(b) The number of birds also decreases (1)
because less food is available to the birds. (1)

(c) (i) Green (1)
When green insects rest on evergreen trees, they are well camouflaged. (1)
The chance of the green insects being seen and predated by the birds is smaller. (1)

(ii) Some insects are green in colour because of genetic variations. (1)
The green insects have a greater chance to survive and reproduce. (1)
Thus the green body colour can be passed on to their offspring. (1)
After many generations, the proportion of green insects becomes greater than those of insects of other colours. (1)

(12 marks)

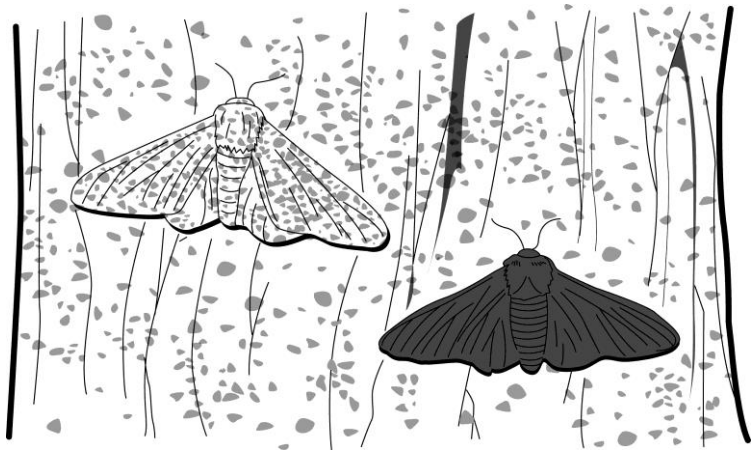
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	Skills assessed	Marks
(a)	Analysing data	3
(b)	Analysing data	1
	Applying concepts	1
(c)(i)	Applying concepts	3
(c)(ii)	Applying concepts	4

|!|BCELB0310000006|!| ★ **Cross topic**

The diagram below shows two varieties of the peppered moth, a dark-bodied moth and a light-bodied moth.



(a) A study was carried out in the 1850s. The numbers of the two peppered moth varieties were investigated in Area A (a rural area) and in Area B (an industrial area). The table below shows the average number of moths per tree.

	Number of black moths	Number of white moths
Area A	0	95
Area B	85	10

The factories in area B released a lot of black smoke and soot. Explain how natural selection might have led to the different distribution of the two varieties of the peppered moth in Area B. (4 marks)

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(b) The colour of the peppered moth is controlled by a pair of alleles. The allele for dark-bodied moths is dominant, while the allele for light-bodied moths is recessive. Using a genetic diagram, determine the proportion of light-bodied moths produced from a cross between two heterozygous dark-bodied moths. (6 marks)

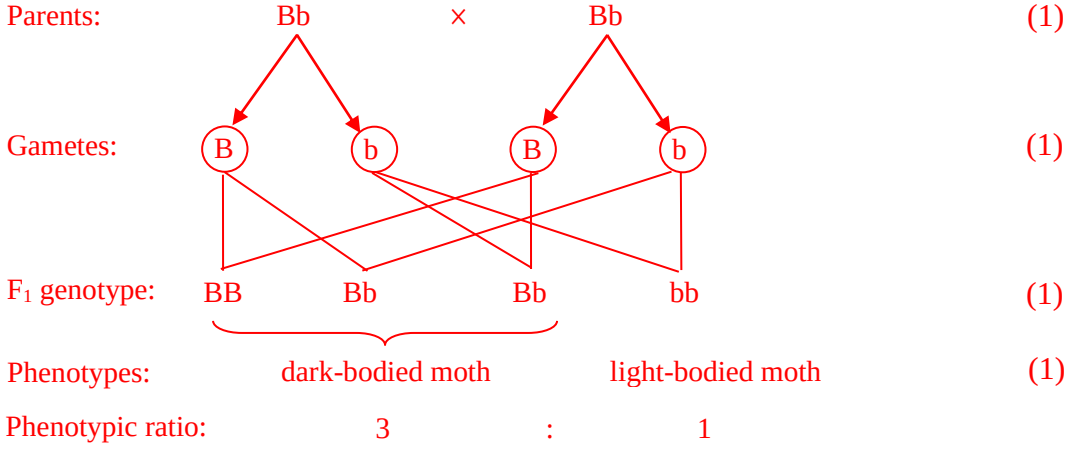
(c) Assuming that a population of peppered moths is separated by a mountain range, describe how a new species of the moths might evolve over a very long period of time. (4 marks)

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##

(a) In Area B (industrial area), the tree trunks were darkened by soot, which provided camouflage to the dark-bodied moths, (1)
while the light-bodied moths were easily seen and eaten by their predators. (1)
The dark-bodied moths had a greater chance to survive and reproduce. (1)
They passed on the allele for the dark body colour to their offspring. Over many generations, the proportion of dark-bodied moths increased in the population. (1)

(b) Let B represent the allele for dark-bodied moths
b represent the allele for light-bodied moths (1)



Proportion of light-bodied moths = 0.25 or $^1/_4$. (1)

(c) Due to geographic isolation by the mountain range, the isolated groups of the moths could not meet or breed with each other. There would be no gene flow between the isolated groups. (1)
The environmental conditions on the two separate areas might be different. (1)
By natural selection, the favourable variations selected for in each isolated group might be different. (1)
Over many generations, different variations might accummulate in each isolated group. The groups of peppered moths might become so different from each other that they could no longer interbreed to produce fertile offspring. (1)

(14 marks)

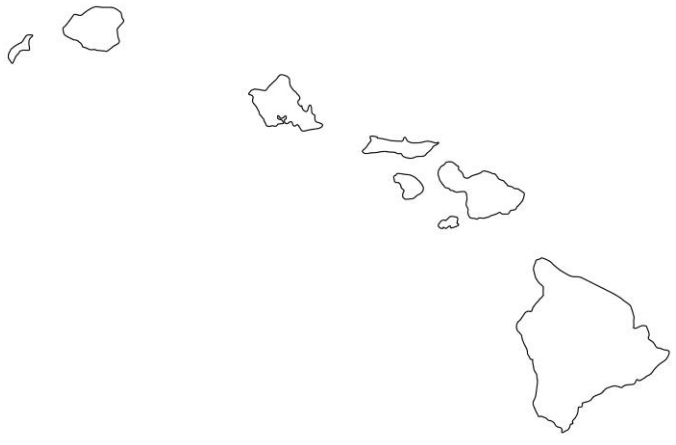
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	Skills assessed	Marks
(a)	Applying concepts	4
(b)	Presenting information	6
(c)	Applying concepts	4

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|!|BCELA0310000007|!| ★★

The diagram below shows the Hawaiian Islands which are 3000 km from the nearest continent. The islands were formed from volcanic activities. Due to the presence of high mountains, local climate varies greatly on each island. 500 species of fruit flies can be found in the Hawaiian Islands.



(a) A large number of fruit fly species have evolved in the Hawaiian Islands. Explain this with the concept of natural selection. (5 marks)

(b) Britain includes many islands. There are about 30 species of fruit flies in Britain but 500 species of fruit flies in the Hawaiian Islands. Based on the degree of isolation of these groups of islands, explain the difference in the number of fruit fly species between Britain and the Hawaiian Islands. (4 marks)

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##

(a) The Hawaiian Islands were separated from each other by the ocean and this geographic barrier separated populations of fruit flies into many isolated groups. (1)
The isolated groups of fruit flies could not meet or breed with each other. (1)
Since the environmental conditions on each island might be different, (1)
different characteristics were selected in each isolated group of fruit flies due to natural selection. (1)
Over many generations, different characteristics accumulated in each isolated group of fruit flies. Different groups of fruit flies became so different from each other that they were unable to interbreed to produce fertile offspring. (1)
New species of fruit flies were formed.

(b) Compared with the Hawaiian Islands, British islands are less isolated / nearer to other continents. (1)
Climate and other environmental conditions in the British islands are less diverse than those in the Hawaiian Islands. (1)
The effect of isolation and envirnomental changes is less strong in the British islands than in the Hawaiian Islands. (1)
The rate of speciation is much slower in the British islands than in the Hawaiian Islands. (1)
So, fewer fruit fly species are found in Britain.

(9 marks)

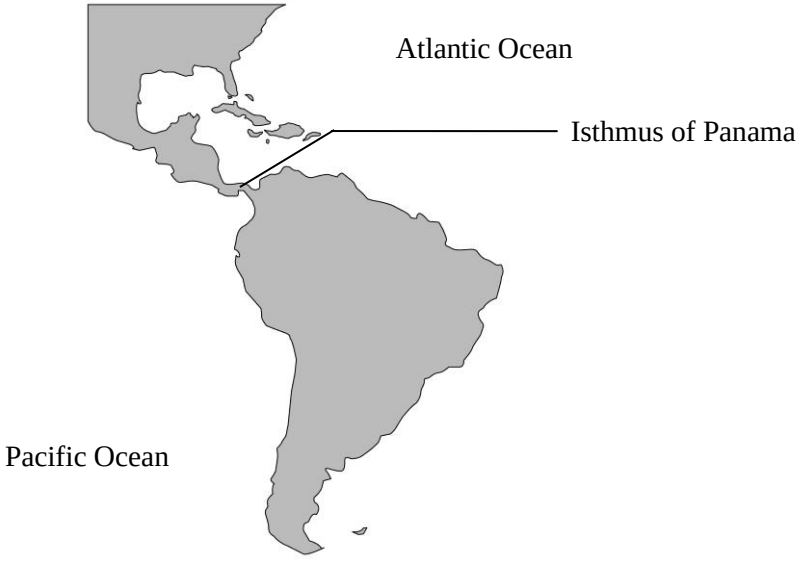
##

	Skills assessed	Marks
(a)	Applying concepts	5
(b)	Applying concepts	4

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|!|BCELA0310000008|!| ★★ **Cross topic**

The Isthmus of Panama is a narrow strip of land that separates the Pacific Ocean from the Atlantic Ocean, linking the North America and South America. The map below shows the Isthmus of Panama and nearby areas.



The photographs below show two species of wrasses.



Cortez rainbow wrasse
(*Thalassoma lucasanum*)



Blue-headed wrasse
(*Thalassoma bifasciatum*)

The Cortez rainbow wrasse is found in the coral reefs on the Pacific side of the isthmus while the blue-headed wrasse is found in the coral reefs on the Atlantic side of the isthmus.
It has been found that both species evolved from a common ancestral population that was split as the isthmus formed.

(a) Which genus do the two types of wrasse belong to? (1 mark)

_____

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(b) How can we determine whether the Cortez rainbow wrasse and the blue-headed wrasse belong to different species? (1 mark)

(c) Suggest how comparative molecular biology can provide evidence to indicate that these wrasse species evolved from a common ancestral population. (2 marks)

(d) Explain how the formation of the isthmus led to the formation of these two separate wrasse species from the common ancestral population. (5 marks)

##

(a) *Thalassoma* (1)

(b) The two types of wrasses are unable to interbreed to produce fertile offspring, so they do not belong to the same species. (1)

(c) When the DNA base sequences of the two wrasse species are compared, (1)
a high similarity in their DNA base sequences indicate that they may have evolved from a common ancestor. (1)

(d) The Atlantic wrasse population and the Pacific wrasse population were geographically separated by the isthmus. (1)
The two populations were unable to breed with each other. (1)

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The environmental conditions are different on the two sides of the isthmus, creating different selection pressures on both sides. (1)

Wrasses that were adapted to the environment can survive and reproduce; certain favourable variations are passed on to the offspring. (1)

Over many generations, different characteristics accumulated in each isolated group of wrasses. The isolated groups of wrasses have become so different that they were unable to interbreed to produce fertile offspring. (1)

(9 marks)

##

	Skills assessed	Marks
(a)	Analysing data	1
(b)	Applying concepts	1
(c)	Applying concepts	2
(d)	Applying concepts	5